Lecture 06B

Linear Maps
Matrix Representation, Change of Basis

MH1201 Linear Algebra 2

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Example. Consider the function $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

To show that $\boldsymbol{T}$ is **additive**, what should we demonstrate?

1. $\boldsymbol{T}(\boldsymbol{A}+\boldsymbol{B}) = \boldsymbol{T}(\boldsymbol{A}) + \boldsymbol{T}(\boldsymbol{B})$ for all matrices $\boldsymbol{A}$ and $\boldsymbol{B}$. ✓
2. $(\boldsymbol{S}+\boldsymbol{T})(\boldsymbol{A}) = \boldsymbol{S}(\boldsymbol{A}) + \boldsymbol{T}(\boldsymbol{A})$ for functions $\boldsymbol{S}$ and $\boldsymbol{T}$.

Recall: $T: V \to W$

- T is additive

if for all $\underset{\sim}{u}, \underset{\sim}{v} \in V$

$T(\underset{\sim}{u} + \underset{\sim}{v})$

$= T(\underset{\sim}{u}) + T(\underset{\sim}{v})$

Example. Consider the function $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

Proof that $\boldsymbol{T}$ is additive

WTS: For all $\hat{A}, \hat{B}$, $\hat{T}(\hat{A}+\hat{B}) = \hat{T}(\hat{A}) + \hat{T}(\hat{B})$ $\hat{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\hat{B} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$

$$\text{LHS} = \hat{T}(\hat{A}+\hat{B}) = \hat{T}\begin{bmatrix} a+\alpha & b+\beta \\ c+\gamma & d+\delta \end{bmatrix} = (a+\alpha+b+\beta, c+\gamma+d+\delta, a+\alpha+c+\gamma, b+\beta+d+\delta)$$

$$\text{RHS} = \hat{T}(\hat{A}) + \hat{T}(\hat{B}) = (a+b, c+d, a+c, b+d) + (\alpha+\beta, \gamma+\delta, \alpha+\gamma, \beta+\delta)$$

$$= (a+b+\alpha+\beta, c+d+\gamma+\delta, a+c+\alpha+\gamma, b+d+\beta+\delta)$$

$\therefore$ LHS = RHS

and so, $\hat{T}$ is additive

Example. Consider the function $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

To show that $\boldsymbol{T}$ is **homogeneous**, what should we demonstrate?

1. $\boldsymbol{T}(\boldsymbol{A}+\boldsymbol{B})=\boldsymbol{T}(\boldsymbol{A})+\boldsymbol{T}(\boldsymbol{B})$ for all matrices $\boldsymbol{A}$ and $\boldsymbol{B}$. (additive)
2. $\boldsymbol{T}(\lambda\boldsymbol{A})=\lambda\boldsymbol{T}(\boldsymbol{A})$ for matrix $\boldsymbol{A}$ and real number λ. (homogeneous)

Example. Consider the function $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

Proof that $\boldsymbol{T}$ is homogeneous

D/Y

Since T is both additive + homogeneous, T is linear

Lecture 06 – Matrix Representation

- Matrix Representation of Linear Maps
  - Finding Range using Matrix Representations
  - Finding Nullspace using Matrix Representations

  } repeat from HBL week 5

  - Standard Matrices
  - Linear Operators
  - Change-of-Basis Matrices

Finding a basis for Range and a basis for Nullspace Using Matrix Representation

Example. Consider the function $T: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

standard basis

What is $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$?

$$T\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = (1,0,1,0)$$

$$T\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = (1,0,0,1)$$

$$T\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = (0,1,1,0)$$

$$T\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = (0,1,0,1)$$

e_1	1	1	0	0
e_2	0	0	1	1
e_3	1	0	1	0
e_4	0	1	0	1
	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

Example. Consider the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$ with

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

What is a basis for range($\boldsymbol{T}$)?

$[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}} =$ HIDDEN $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

Is $[(1,0,1,0), (1,0,0,1), (0,1,1,0), (0,1,0,1)]$ a basis for range(T)?

No. We only know $[\ldots]$ span range(T). We still need to check linear independence.

↳ i.e. we row reduce $[T]_{\mathcal{B}}^{\mathcal{A}}$

Example.

Consider the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

What is a basis for range$(\boldsymbol{T})$?

$[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}} =$ HIDDEN

$$\begin{array}{cccc} & & & \\ \left[\begin{matrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{matrix}\right] & \xRightarrow{\text{rref}} & \left[\begin{matrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}\right] \\ \begin{matrix} \Delta & \Delta & \Delta & \\ E_{11} & E_{12} & E_{21} & \end{matrix} & & \begin{matrix} \Delta & \Delta & \Delta & \end{matrix} \end{array}$$

A basis would be $[T(E_{11}), T(E_{12}), T(E_{21})]$

$= [(1,0,1,0), (1,0,0,1), (0,1,1,0)]$

E_{22} does NOT appear means that E_{22} belongs to span(E_{11}, E_{12}, E_{21}).

Example.

Consider the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

Is $\boldsymbol{T}$ surjective? No,

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Recall: Proposition.

Let $\boldsymbol{T}: \mathcal{V} \to \mathcal{W}$ be a linear map.

$\boldsymbol{T}$ is surjective if and only if $\dim \operatorname{range}(\boldsymbol{T}) = \dim \mathcal{W}$.

from previous slide

$\dim \operatorname{range}(T) = 3$ $\quad$ $\dim W = 4$

Since $\dim \operatorname{range}(T) \neq \dim(\text{codomain})$, T is NOT surjective

Example.

Consider the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

What is a basis for null $(\boldsymbol{T})$?

$[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}} =$ HIDDEN

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \xrightarrow{\text{rref}} \begin{array}{cccc} a & b & c & \\ \left[\begin{matrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}\right] \\ \Delta & \Delta & \Delta & t \end{array}$$

t ↓ free Variable

$$\text{null}\left([T]_{\mathcal{B}}^{\mathcal{A}}\right) = \left\{ \begin{bmatrix} a \\ b \\ c \\ t \end{bmatrix} : \begin{matrix} a = t \\ b = -t \\ c = -t \end{matrix}, \ t \in \mathbb{R} \right\}$$

$$= \left\{ t \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \ t \in \mathbb{R} \right\}$$

$$= \text{span}\left(\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right)$$ → can just conclude immediately to here

Is $\left(\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}\right)$ a basis for null space of T? No.

→ nullspace of linear map should be subspace of domain → so the ans should be a matrix.

$$\begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \begin{matrix} E_{11} \\ E_{12} \\ E_{21} \\ E_{22} \end{matrix}$$

$E_{11} - E_{12} - E_{21} + E_{22} \Rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ basis for null(T).

Example.

Consider the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4$

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b, c+d, a+c, b+d).$$

Consider the basis $\mathcal{B} = [\boldsymbol{E}_{11}, \boldsymbol{E}_{12}, \boldsymbol{E}_{21}, \boldsymbol{E}_{22}]$ for $\mathcal{M}_{2\times 2}(\mathbb{R})$ and $\mathcal{A} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4]$ for $\mathbb{R}^4$.

Is $\boldsymbol{T}$ ~~surjective~~? No.

injective

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Recall:

Proposition. (Conditions for Injectivity)

Let $\boldsymbol{T}: \mathcal{V} \to \mathcal{W}$ be a linear map.

$\boldsymbol{T}$ is injective if and only if $\dim \operatorname{null}(\boldsymbol{T}) = 0$.

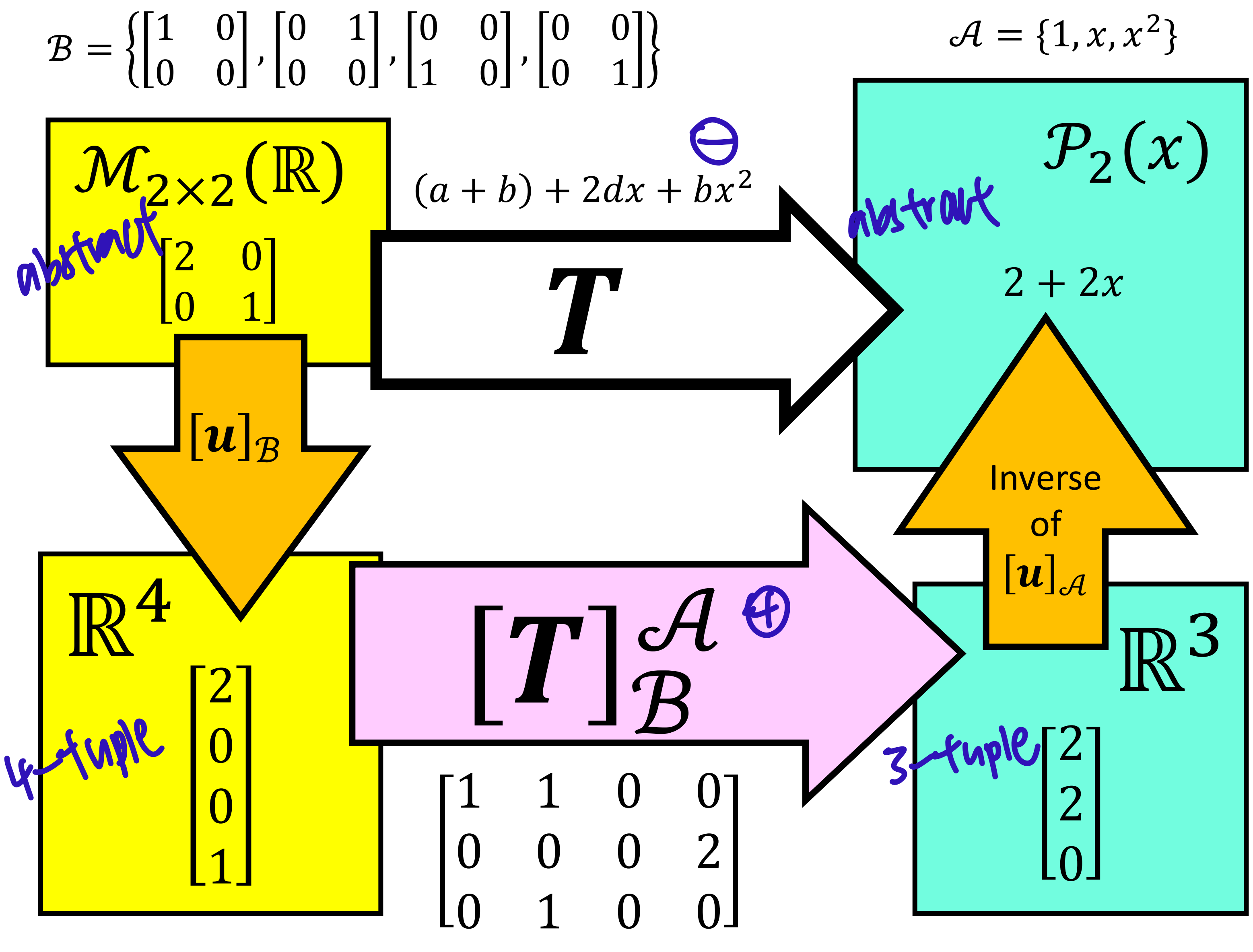
$\dim \operatorname{null}(T) = 1$

Since $\dim \operatorname{null}(T) \neq 0$, T is NOT injective.

Matrix Representation

- Standard Matrices (special matrix rep)
- Linear Operators (special linear map)
- Change-of-Basis Matrices (special matrix rep)

Matrix Representation of Linear Maps



Definition. Let $\boldsymbol{T} : \mathcal{V} \to \mathcal{W}$ be a linear map.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ be a basis for $\mathcal{V}$.

Let $\mathcal{A} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_m\}$ be a basis for $\mathcal{W}$.

Then the *matrix representation with respect to bases $\mathcal{A}$ and $\mathcal{B}$*, denoted by $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$, is the matrix such that

$$\boldsymbol{T}(\boldsymbol{v}_j) = \boldsymbol{T}_{1,j}\boldsymbol{w}_1 + \cdots + \boldsymbol{T}_{m,j}\boldsymbol{w}_m.$$

vector space $\xrightarrow{\text{linear}}$ vector space

$\mathbb{R}^m$ $\mathbb{R}^n$

⊖ Not many libraries have built in functions

⊕ many libraries tailored from matrix manipulation

Matrix Representation of Linear Maps

$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

$\mathcal{A} = \{1, x, x^2\}$

$$\begin{tikzcd}
\mathcal{M}_{2\times 2}(\mathbb{R}) \ni \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \underset{\sim}{u} \arrow[r, "T"', "(a+b)+2dx+bx^2"] \arrow[d, "{[\boldsymbol{u}]_{\mathcal{B}}}"] & \mathcal{P}_2(x) \ni 2+2x = T(\underset{\sim}{u}) \arrow[d, "{[\boldsymbol{u}]_{\mathcal{A}}}"] \\
\mathbb{R}^4 \ni \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix} = [\underset{\sim}{u}]_{\mathcal{B}} \arrow[r, "{[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}}", "{\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}}"'] & \mathbb{R}^3 \ni \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} = [T(\underset{\sim}{u})]_{\mathcal{A}}
\end{tikzcd}$$

$[T]_{\mathcal{B}}^{\mathcal{A}} [\underset{\sim}{u}]_{\mathcal{B}}$

Proposition.

$$[\boldsymbol{T}(\boldsymbol{u})]_{\mathcal{A}} = [\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}} [\boldsymbol{u}]_{\mathcal{B}}.$$

fancy description

$\Leftrightarrow$ this diagram commutes

Typical in Algebra course

Definition. Let $\boldsymbol{T} : \mathcal{V} \to \mathcal{W}$ be a linear map.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ be a basis for $\mathcal{V}$ and $\mathcal{A} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_m\}$ be a basis for $\mathcal{W}$.

Then the *matrix representation with respect to bases $\mathcal{A}$ and $\mathcal{B}$*, denoted by $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$, is the matrix such that

$$\boldsymbol{T}(\boldsymbol{v}_j) = \boldsymbol{T}_{1,j}\boldsymbol{w}_1 + \cdots + \boldsymbol{T}_{m,j}\boldsymbol{w}_m.$$

$[T]_{\mathcal{B}}^{\mathcal{A}}$ basis

When $\mathcal{A}$ and $\mathcal{B}$ are standard bases, we refer to $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{A}}$ as the ***standard matrix***.

Example

Define the linear map $\boldsymbol{T}: \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ such that

$$\boldsymbol{T}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = (a+b) + 2dx + bx^2.$$

Then the standard matrix for $\boldsymbol{T}$ is $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}$, because

$\mathcal{A} = [1, x, x^2]$

$\mathcal{B} = [E_{11}, E_{12}, E_{21}, E_{22}]$

are both standard bases.

Definition. A ***linear operator*** is a linear map $\boldsymbol{T} : \mathcal{V} \to \mathcal{V}$ whose domain and codomain are the same.

If we choose the same basis for domain and codomain, then $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a square matrix.

Example

Define the linear map $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then $\boldsymbol{T}$ is a linear operator, because

$\mathbb{R}^2$ is both the domain and the codomain.

Example

Define the linear operator $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Find the standard matrix for $\boldsymbol{T}$.

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standard basis for $\mathbb{R}^2 = \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right] = \mathcal{B}$

$$T\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 41 \\ -12 \end{bmatrix} = 41 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-12)\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -12 \\ 34 \end{bmatrix} = -12\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 34\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{standard matrix} = \left[T \right]_{\mathcal{B}}^{\mathcal{B}} = \begin{array}{c} \\ \\ \end{array}\begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{array}{c} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array}$$

not a coincidence that it is same as given in qn wooclap

Example

Define the linear operator $\boldsymbol{T}\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that

basis different

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

matrix same

Let $\mathcal{A} = \left[\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\right]$.

Find the matrix representation $[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}}$.

$$T\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} = \begin{bmatrix} -40 \\ 30 \end{bmatrix} = 50\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} + 0\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$T\begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} = \begin{bmatrix} 15 \\ 20 \end{bmatrix} = 0\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} + 25\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$[T]_{\mathcal{A}}^{\mathcal{A}} = \begin{array}{c} \begin{bmatrix} 50 & 0 \\ 0 & 25 \end{bmatrix} \begin{array}{c} \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} \\ \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \end{array} \\ \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} \quad \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \end{array}$$

Why Change Basis?

Example

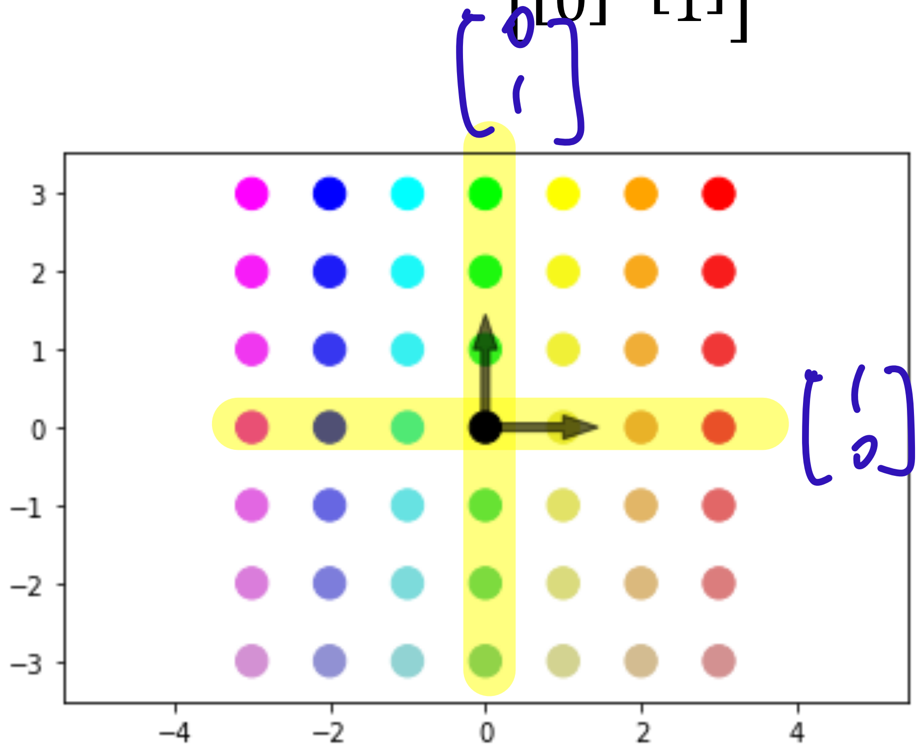
$$T\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}41 & -12\\-12 & 34\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}.$$

Domain: $\mathbb{R}^2$

Basis: $\mathcal{B}=\left[\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right]$

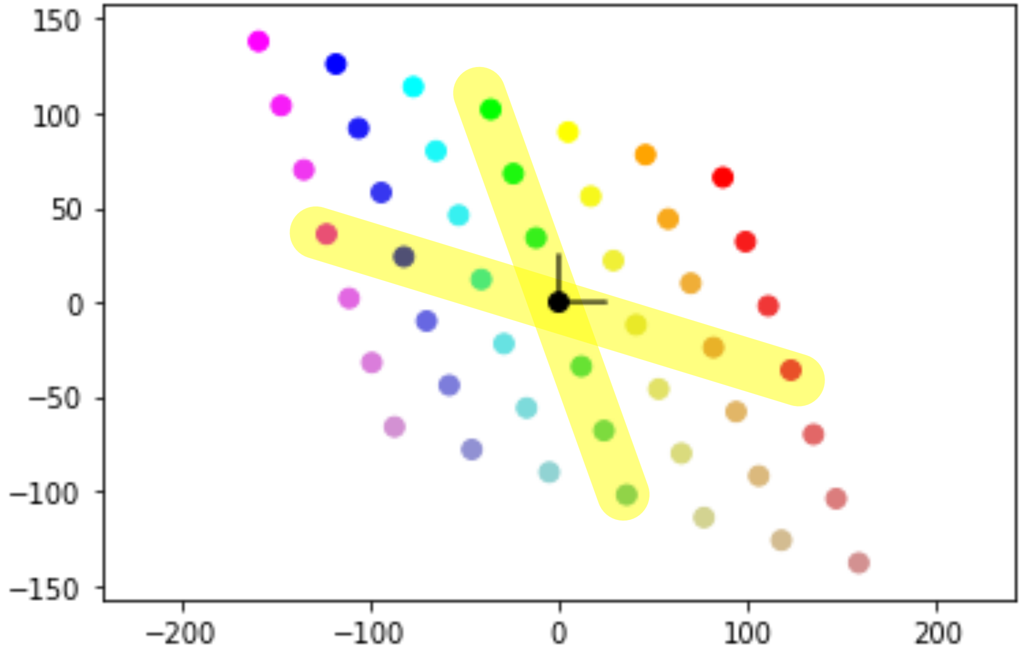
Codomain: $\mathbb{R}^2$

Basis: $\mathcal{B}=\left[\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right]$



$\xrightarrow{T}$

observe that the axes was rotated.



Why Change Basis?

Example

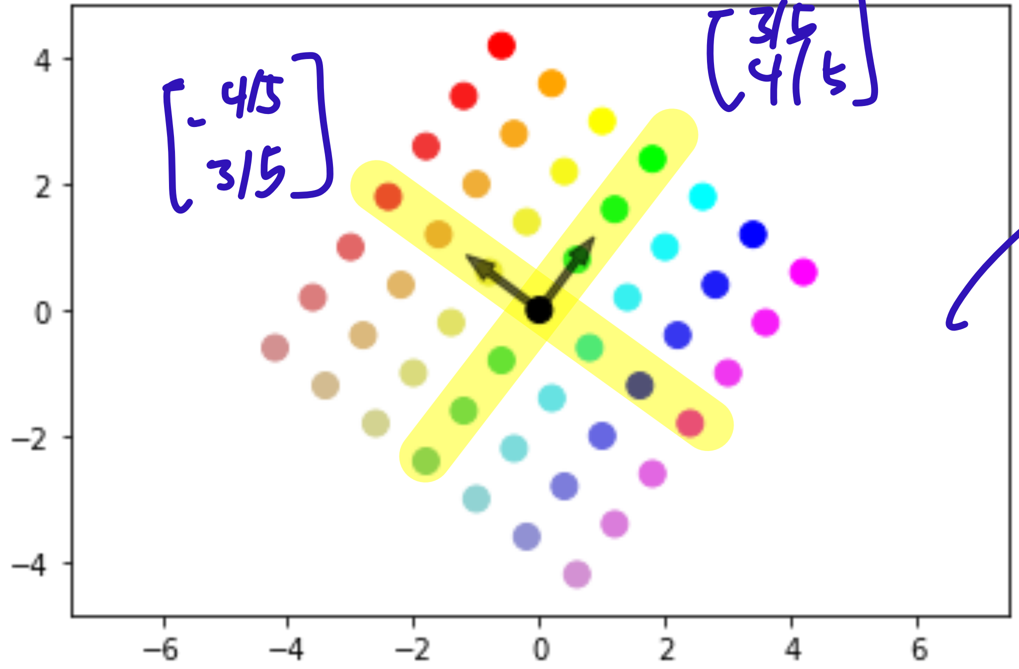
$$T\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}41 & -12\\-12 & 34\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}.$$

Domain: $\mathbb{R}^2$

Basis: $\mathcal{A} = \left[\begin{bmatrix}-4/5\\3/5\end{bmatrix}, \begin{bmatrix}3/5\\4/5\end{bmatrix}\right]$

Codomain: $\mathbb{R}^2$

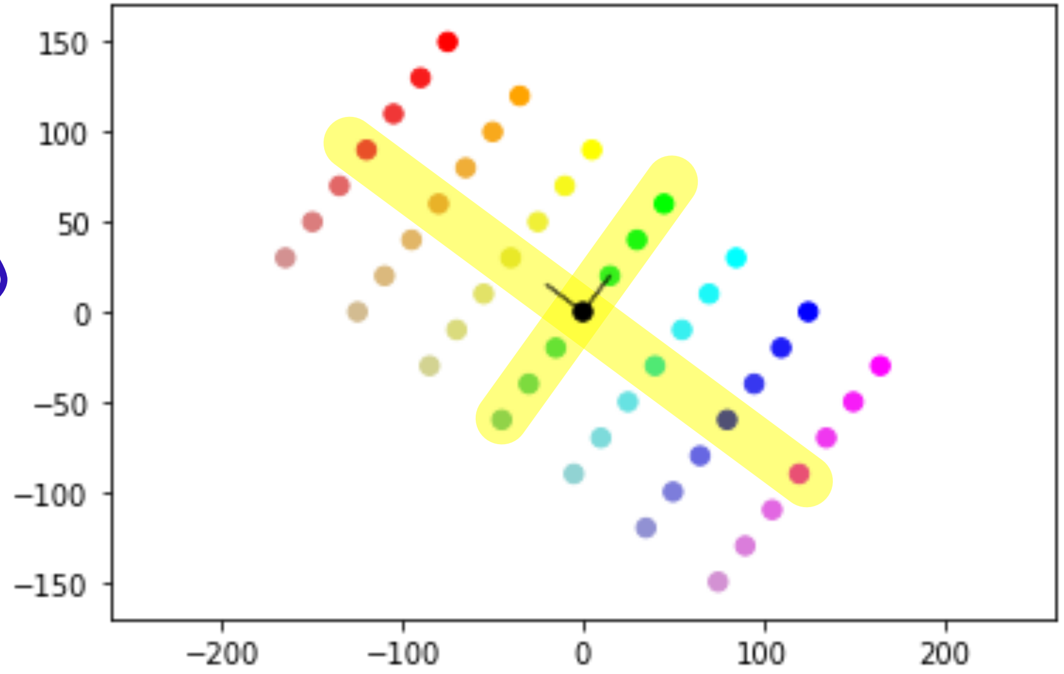
Basis: $\mathcal{A} = \left[\begin{bmatrix}-4/5\\3/5\end{bmatrix}, \begin{bmatrix}3/5\\4/5\end{bmatrix}\right]$



$\begin{bmatrix}-4/5\\3/5\end{bmatrix}$ $\begin{bmatrix}3/5\\4/5\end{bmatrix}$

T

axes do not rotate



Example

Define the linear operator $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

Let $\mathcal{A} = \left[\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\right]$.

Find the matrix representation $[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}}$ using the fact

$$[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}} = [\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}}$$

changing basis

let's try to find

$[T]_{\mathcal{A}}^{\mathcal{A}}$

$\underset{\sim}{x} = T(\underset{\sim}{v})$

$$\begin{array}{ccccccc} \mathbb{R}^2 & & \mathbb{R}^2 & \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix} & \mathbb{R}^2 & & \mathbb{R}^2 \\ \boxed{\underset{\sim}{v}} & \xrightarrow{I} & \boxed{\underset{\sim}{v}} & \xrightarrow{T} & \boxed{\underset{\sim}{v}} & \xrightarrow{I} & \boxed{\underset{\sim}{v}} \\ \mathcal{A} & & \mathcal{B} & & \mathcal{B} & & \mathcal{A} \end{array}$$

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Example

Define the linear operator $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

Let $\mathcal{A} = \left[\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\right]$ and $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right]$.

Find the change-of-basis matrix $[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}}$.

Definition.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ be a basis for $\mathcal{V}$.

Let $\mathcal{A} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_n\}$ be another basis for $\mathcal{V}$.

Then the *change-of-basis matrix from basis $\mathcal{B}$ to basis $\mathcal{A}$* is the matrix $[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}$.

Here, $\boldsymbol{I}$ is the identity map.

get back itself

$$[I]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix} \begin{matrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} \quad \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$I\begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix} = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}$$

$$= -4/5\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3/5\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

DIY (second column)

Example

Define the linear operator $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$\boldsymbol{T}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Let $\mathcal{A} = \left[\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\right]$ and $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right]$.

Find the change-of-basis matrix $[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}$.

wooclap

Definition.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be a basis for $\mathcal{V}$.

Let $\mathcal{A} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n\}$ be another basis for $\mathcal{V}$.

Then the *change-of-basis matrix from basis $\mathcal{B}$ to basis $\mathcal{A}$* is the matrix $[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}$.

Here, $\boldsymbol{I}$ is the identity map.

$$[I]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}$ $\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = -4/5 \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} + 3/5 \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3/5 \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} + 4/5 \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

Example Define the linear operator $\boldsymbol{T}: \mathbb{R}^2 \to \mathbb{R}^2$ such that

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Let $\mathcal{A} = \left[\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}\right]$ and $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right]$.

Find the matrix representation $[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}}$ using the fact

$$[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}} = [\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}}$$

wooclap

$$[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}$$

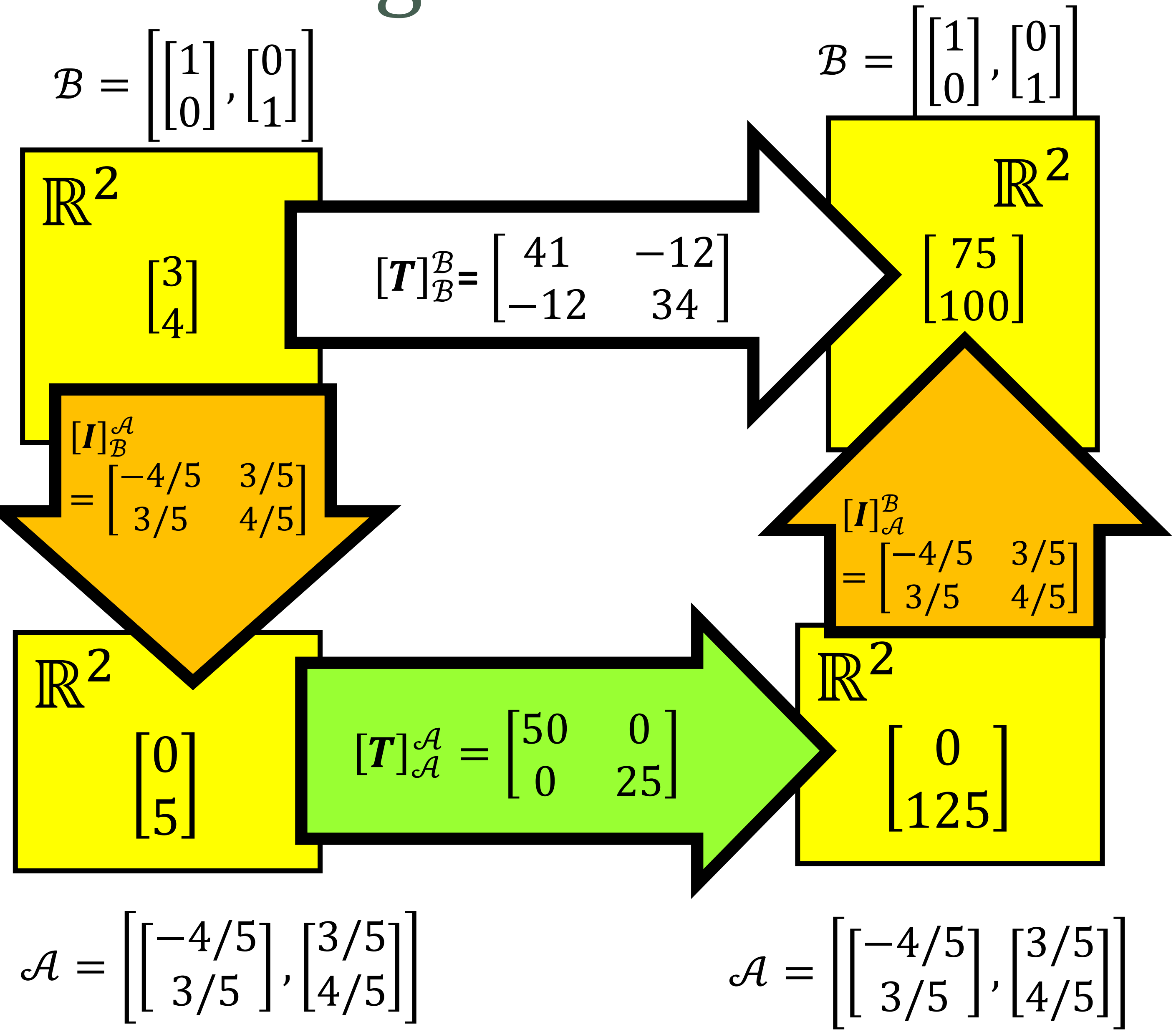
$$[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}$$

$$[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}$$

$$\begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}\underbrace{\begin{bmatrix} 41 & -12 \\ -12 & 34 \end{bmatrix}\begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}}_{\text{multiply this first}} = \begin{bmatrix} -4/5 & 3/5 \\ 3/5 & 4/5 \end{bmatrix}\begin{bmatrix} -40 & 15 \\ 30 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} 50 & 0 \\ 0 & 25 \end{bmatrix}$$

Change-of-Basis

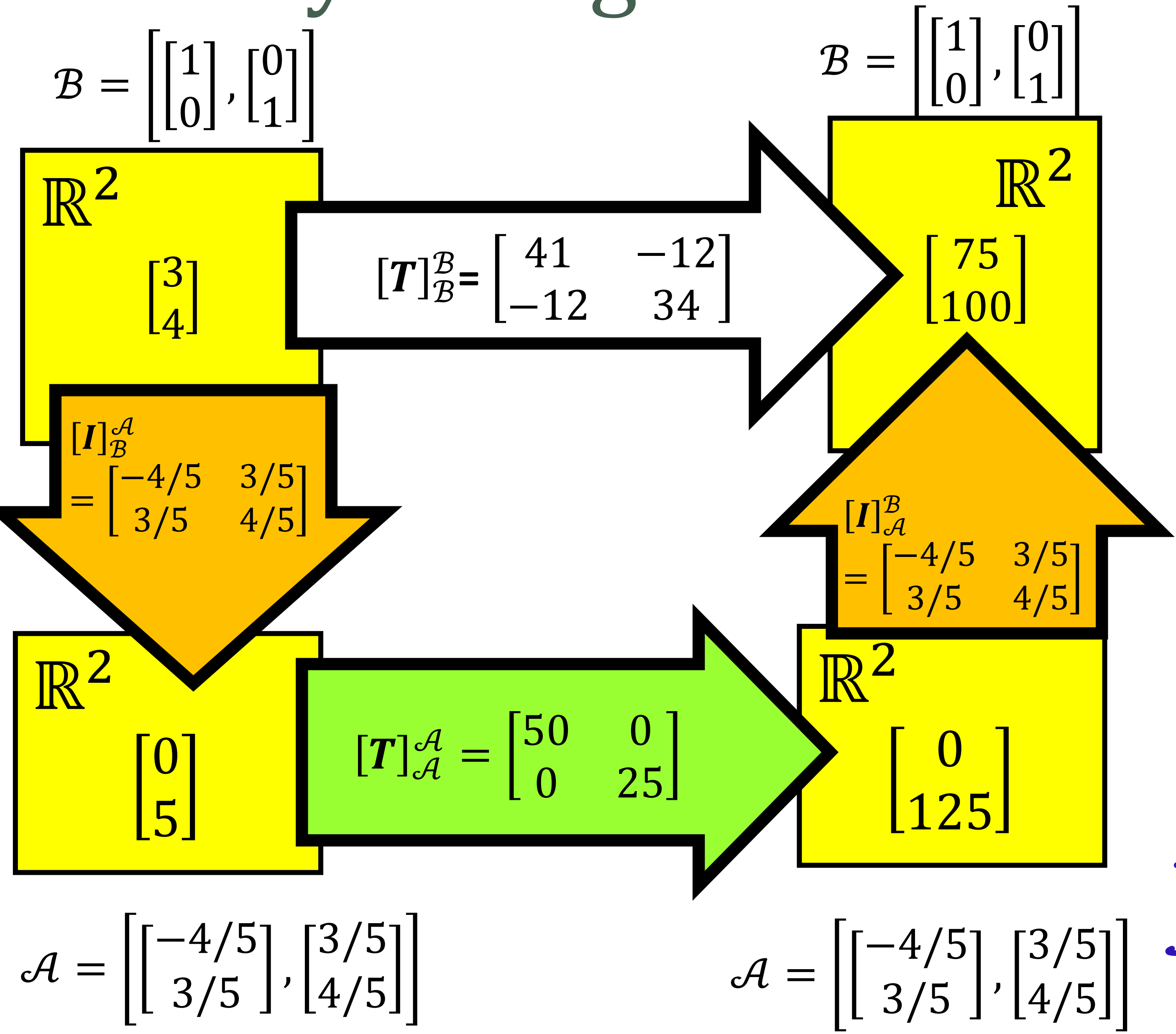


Proposition.

$$[T]_{\mathcal{A}}^{\mathcal{A}} = [I]_{\mathcal{B}}^{\mathcal{A}}[T]_{\mathcal{B}}^{\mathcal{B}}[I]_{\mathcal{A}}^{\mathcal{B}}.$$

Proof omitted

Why Change Basis?



- Multiplying by $[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}}$ is easier than multiplying by $[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}}$

Multiplying $\begin{bmatrix}41 & -12\\-12 & 34\end{bmatrix}\begin{bmatrix}3\\4\end{bmatrix} = \begin{bmatrix}75\\100\end{bmatrix}$

is harder

than multiplying $\begin{bmatrix}50 & 0\\0 & 25\end{bmatrix}\begin{bmatrix}0\\5\end{bmatrix}$

$= \begin{bmatrix}0\\125\end{bmatrix}$

Question: How to find $\mathcal{A}$?

$\rightarrow$ the vectors in $\mathcal{A}$ are eigenvectors

$\rightarrow$ process: Diagonalization.

$$
\begin{tikzcd}[column sep=huge, row sep=huge]
\overset{\mathcal{B}=\left[\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right]}{\mathbb{R}^2 \ni \begin{bmatrix}3\\4\end{bmatrix}} \arrow[r, "{[\boldsymbol{T}]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix}41 & -12\\ -12 & 34\end{bmatrix}}"] \arrow[d, "{[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}} = \begin{bmatrix}-4/5 & 3/5\\ 3/5 & 4/5\end{bmatrix}}"', "\hat{M}"] & \overset{\mathcal{B}=\left[\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right]}{\mathbb{R}^2 \ni \begin{bmatrix}75\\100\end{bmatrix}} \\
\underset{\mathcal{A}=\left[\begin{bmatrix}-4/5\\3/5\end{bmatrix},\begin{bmatrix}3/5\\4/5\end{bmatrix}\right]}{\mathbb{R}^2 \ni \begin{bmatrix}0\\5\end{bmatrix}} \arrow[r, "{[\boldsymbol{T}]_{\mathcal{A}}^{\mathcal{A}} = \begin{bmatrix}50 & 0\\ 0 & 25\end{bmatrix}}"] & \underset{\mathcal{A}=\left[\begin{bmatrix}-4/5\\3/5\end{bmatrix},\begin{bmatrix}3/5\\4/5\end{bmatrix}\right]}{\mathbb{R}^2 \ni \begin{bmatrix}0\\125\end{bmatrix}} \arrow[u, "{[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix}-4/5 & 3/5\\ 3/5 & 4/5\end{bmatrix}}"', "\hat{M}^{-1}"]
\end{tikzcd}
$$

Proposition.

$[\boldsymbol{I}]_{\mathcal{A}}^{\mathcal{B}}$ is the inverse of $[\boldsymbol{I}]_{\mathcal{B}}^{\mathcal{A}}$.

Proof Omitted.

Notice that $\hat{M} = \hat{M}^{-1}$

We will study matrices

$\hat{M}^{-1} = \hat{M}^{T}$

↳ orthogonal matrices

After Term Break

- Week 8, 9
  - Eigenvalues and Eigenvectors
  - Diagonalization
- Week 10, 11
  - Inner Product Space
  - Orthonormal Basis
  - Gram-Schmidt Process
- Week 12, 13
  - Revision